

The Digital Wave

ARPANET was transformed into the Internet as we know it today. The technology involved in this was a very basic packet switching network that later moved to include packet satellite networks, ground-based packet radio networks and other networks. The basic idea behind using the Internet was connecting multiple independent networks of rather arbitrary design, beginning with the ARPANET that, in other terms, follows a technical idea of open architecture networking.

Until that time, there was only one method of networking, which was the circuit switching method. In this method, networks interconnecting at the circuit level passed individual bits on a synchronous basis along a portion of an end-to-end circuit between a pair of end locations.

What this means, is that in the circuit switching mode, there is a dedicated path by which data travels from the source to the destination. In packet switching mode, this is not the case. The hub or the router, which acts as the intermediary, decides the best path for sending or receiving the data packets that gives this process more flexibility with increased efficiency.

The future of networking was based on this skeleton. Further down the line, the TCP/IP (Transmission Control Protocol/Internet Protocol) was developed since the Network Control Protocol of NCP, which was in use at that time, did not have the ability to address networks (and machines) further downstream than a destination IMP. This was the main driving force behind the development of another protocol which we know as TCP/IP, today.

Further, down-the-line applications such as Telnet were developed, which is a very basic application for remote logging and file transfer. However, e-mail remained the biggest and single most innovation from that era. Other applications being developed at that time included packet-based voice communication (the precursor of Internet telephony), various models of file and disk shar-